

Document Extraction on Low-Cost Cloud GPUs: Speed, Quality and Cost

We ran an open-source language model on rented cloud GPUs to pull structured information out of five kinds of business documents. We varied the hardware and the batch size across four experiments and measured speed, accuracy and cost.

Data: every result uses the public Hugging Face dataset `Lucius-Morningstar/mailroom-dataset`. No American Family Insurance data was used or shared.

Key findings

- 1 **Adding a second GPU doubles speed at the same cost per document.** On the same 250 documents, moving from 1 GPU to 2 GPUs processed 99% more documents per minute, and GPU cost per document changed by only +0.4%. The trade-off: each document waited 1.4–1.8 times longer for its answer, because more documents shared the GPUs at once. This setup suits batch processing; latency-sensitive use would need different tuning.
- 2 **Larger batches are cheaper.** On the 2-GPU setup, processing 100 documents of each type instead of 50 cut GPU cost per document by 19% (four document types; merger agreements excluded because their settings changed). 1 of 400 documents failed.
- 3 **Hardware choice did not change accuracy.** Comparing the same documents across experiments (420 document pairs), the average score change per document type was between -0.017 and +0.023 on a 0–1 scale, and every 95% confidence interval includes zero. Effects smaller than about ± 0.04 cannot be ruled out at this sample size.
- 4 **Merger agreements are the weak spot; reading the whole agreement helps.** These contracts are far longer than the model can read in one pass. Splitting each agreement into overlapping sections and combining the answers raised accuracy from 3.5% to 14.0% and the share of questions answered from 23% to 69%, at 4.5 times the cost per agreement.
- 5 **The full study cost \$3.39 in cloud charges for 1,050 documents** (about 0.32 cents each, all-in), and \$0.00 was billed after the provider's free credits.

What we tested

The task. An automated mailroom reads each incoming document and returns its key facts as structured data (for example parties, dates, amounts, and which contract clauses are present). Each of the five document types has its own extraction instructions:

Document type	What it contains	How accuracy is measured
Insurance Claims	Insurance claim forms and notices	Extraction score
Contracts	Commercial contracts	Clause F1 (CUAD)
Corporate Records	Corporate filings and records	Extraction score
Correspondence	Business letters and email	Extraction score
Merger Agreements	Merger agreements (often 100+ pages)	Question accuracy (MAUD)

The model. Qwen/Qwen3-8B-AWQ: an open-weights model with 8 billion parameters, compressed to 4-bit weights (AWQ) so it fits on a single 24 GB GPU. It ran on vLLM, an open-source model server, on NVIDIA L4 GPUs rented from Modal at \$0.80 per GPU-hour.

The sample. Documents were drawn at random with a fixed seed (42), so every run is reproducible. Smaller samples are subsets of larger ones, so runs can be compared on the same documents.

The four experiments

Experiment	GPUs	Documents processed at once	Documents per type	Total documents
Experiment 1	1	8	20	100
Experiment 2	1	8	50	250
Experiment 3	2	32	50	250
Experiment 4	2	32	100 (merger agreements: 50)	450

Experiments 2 and 3 use the identical 250 documents and differ only in GPU count and documents processed at once, so they give the cleanest hardware comparison. Experiment 4 repeats Experiment 3's hardware on twice as many documents and also tests improved settings for merger agreements (described below).

How to read the numbers

Term	Meaning
Token	The unit a language model reads and writes; roughly three-quarters of an English word.
Documents per minute	Overall throughput for the run: documents finished ÷ total run time.
Tokens per second per GPU	Text processed (read plus written) per second by each GPU; a hardware-efficiency measure.

Term	Meaning
Median time per document	Half of documents finished faster than this, half slower (seconds from submission to answer).
GPU cost per document	GPU time the documents actually used × \$0.80/hour ÷ documents. Excludes start-up and idle time.
Session cost	What the cloud provider metered for the whole session, including start-up, waiting and shut-down.
Failed document	The model's answer exceeded the output length limit (8,192 tokens) and was cut off, so no result was returned.
Extraction score	0–1. Average agreement between extracted fields and the dataset's answer key, per document.
Clause F1 (CUAD)	0–1. CUAD is a public set of commercial contracts with lawyer-labeled clause types. F1 balances finding the clauses that are present against flagging clauses that are not.
Question accuracy (MAUD)	0–1. MAUD is a public set of merger agreements with lawyer-written multiple-choice questions. Accuracy is the share of all labeled questions answered correctly; coverage is the share the model answered at all.
95% confidence interval	The range that likely contains the true difference; if it includes zero, the data show no reliable difference.

The three scores use different scales, so compare them across experiments, not across document types.

Speed and cost by experiment

Measure	Experiment 1	Experiment 2	Experiment 3	Experiment 4
Documents per minute	8.7	10.4	20.7	11.9
Tokens per second per GPU	812	1,002	1,013	1,662
GPU cost per document	\$0.00153	\$0.00128	\$0.00129	\$0.00225
Failed documents	3 of 100	7 of 250	5 of 250	1 of 450

Every failure was an answer cut off at the output length limit, all on contracts or merger agreements (the longest documents).

Experiment 4's cost and throughput include the slower, more thorough merger-agreement settings; the like-for-like batch-size comparison is in Key finding 2.

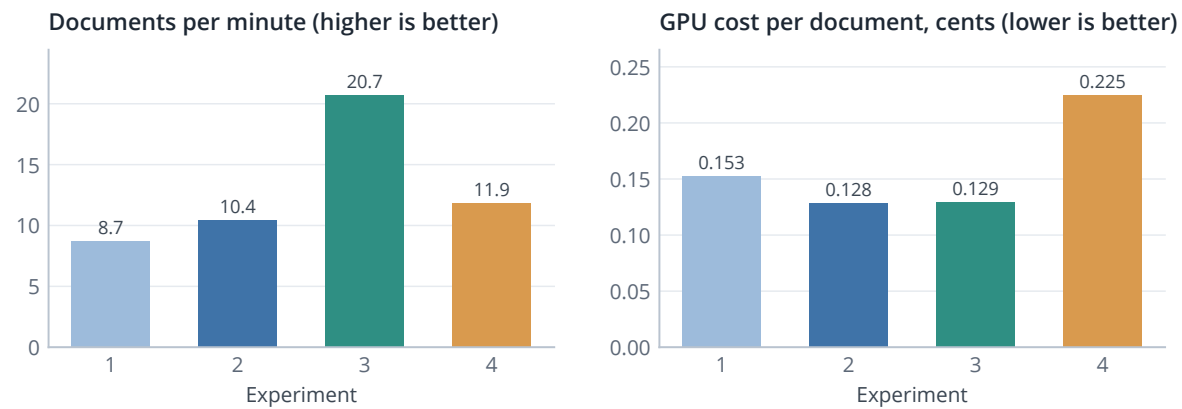


Figure 1. Throughput and GPU cost per document for each experiment.

Results by document type

Scores by document type (0–1, higher is better):

Document type	Measure	Experiment 1	Experiment 2	Experiment 3	Experiment 4
Insurance Claims	Extraction score	0.684	0.684	0.686	0.672
Contracts	Clause F1 (CUAD)	0.597	0.590	0.605	0.608
Corporate Records	Extraction score	0.459	0.449	0.452	0.475
Correspondence	Extraction score	0.327	0.345	0.334	0.341
Merger Agreements	Question accuracy (MAUD)	0.014	0.048	0.035	0.140*

* Experiment 4 uses the improved merger-agreement settings.

Cost per successfully processed document (US dollars):

Document type	Experiment 1	Experiment 2	Experiment 3	Experiment 4
Insurance Claims	\$0.00042	\$0.00039	\$0.00032	\$0.00036
Contracts	\$0.00350	\$0.00310	\$0.00284	\$0.00207
Corporate Records	\$0.00026	\$0.00032	\$0.00019	\$0.00021
Correspondence	\$0.00011	\$0.00018	\$0.00012	\$0.00012
Merger Agreements	\$0.00390	\$0.00284	\$0.00329	\$0.01475

Median seconds per document:

Document type	Experiment 1	Experiment 2	Experiment 3	Experiment 4
Insurance Claims	13	14	20	20
Contracts	66	68	103	96
Corporate Records	14	13	24	20
Correspondence	6	7	11	10
Merger Agreements	50	51	92	1,044

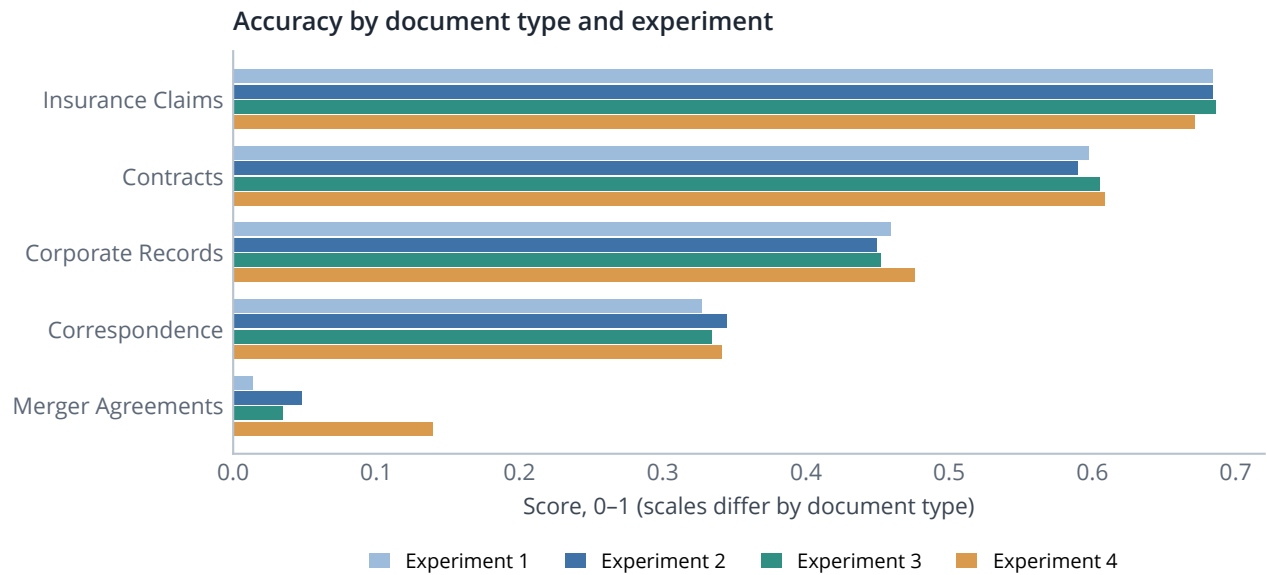


Figure 2. Accuracy by document type and experiment. Each document type has its own scale; compare bars within a type, not across types.

Merger agreements: what changed

Merger agreements are long: the median agreement in the sample runs to hundreds of thousands of characters, while the standard settings read only the first and last 30,000 characters combined. In Experiment 4 we changed the settings for this document type only, on the same 50 agreements as Experiment 3:

Setting	Standard (Experiments 1–3)	Improved (Experiment 4)
What the model reads	Start and end of the agreement; the middle is skipped	The whole agreement, in overlapping sections of about 47,000 characters, with answers combined
Instructions	General extraction instructions	Instructions written around the MAUD question set
Output length limit	8,192 tokens	6,144 tokens, with one retry if an answer is cut off
Sampling	temperature 0.7, other sampling at vLLM defaults	temperature 0.7, top_p 0.8, top_k 20, presence_penalty 1.0
Question accuracy	3.5%	14.0%
Questions answered	23%	69%
Agreements returning a result	46 of 50	50 of 50
Cost per agreement	\$0.0033	\$0.0147
Median time per agreement	92 s	1,044 s

On the 46 agreements scored under both settings, 35 improved and 1 got worse (average change +0.106 on the 0–1 scale). Several settings changed together, so this run does not show which change drove the gain.

Total cost

GPU cost counts only the time the GPUs spent on documents. Session cost is the provider's metered charge for the whole session, including start-up, waiting between runs and shut-down.

Experiments	Documents	GPU cost on documents	Session cost	Share of session spent on documents	Session cost per document	Billed
Experiments 1 and 3	350	\$0.47	\$1.09	44%	\$0.0031	\$0.00
Experiment 2	250	\$0.32	\$0.49	65%	\$0.0020	\$0.00
Experiment 4	450	\$1.01	\$1.81	56%	\$0.0040	\$0.00
All experiments	1,050	\$1.81	\$3.39	53%	\$0.0032	\$0.00

Billed is zero because the provider's monthly free credits covered the charges. Session costs exclude short exploratory test deployments run between the main runs.

Limitations

- **One sample per run.** Each run used one random draw of 20–100 documents per type; small differences between experiments may be noise.
- **Hardware and load changed together.** Going from 1 to 2 GPUs also raised the number of documents processed at once (8 to 32), so the speed gain cannot be split between the two.
- **Some outputs vary between runs.** Contracts and merger agreements are generated with some randomness (temperature 0.7), so repeated runs give slightly different answers.
- **Strict format checks.** Most insurance-claim outputs did not pass strict data-format validation even though their extraction scores are the highest; a system that rejects malformed records would need that fixed first.
- **Public data only.** Results describe this public dataset; performance on other document collections has not been measured.

Experiment cross-reference

For readers comparing against the project's internal records:

Experiment	Internal shorthand
Experiment 1	1×L4 C8 n=20
Experiment 2	1×L4 C8 n=50
Experiment 3	2×L4 C32 n=50
Experiment 4	2×L4 C32 n=100

Shorthand key: 2×L4 = two NVIDIA L4 GPUs; C32 = 32 documents processed at once; n=100 = documents per type. Detailed tables and charts: [SAND-37-MASTER-APPENDIX.md](#) in the project repository.